

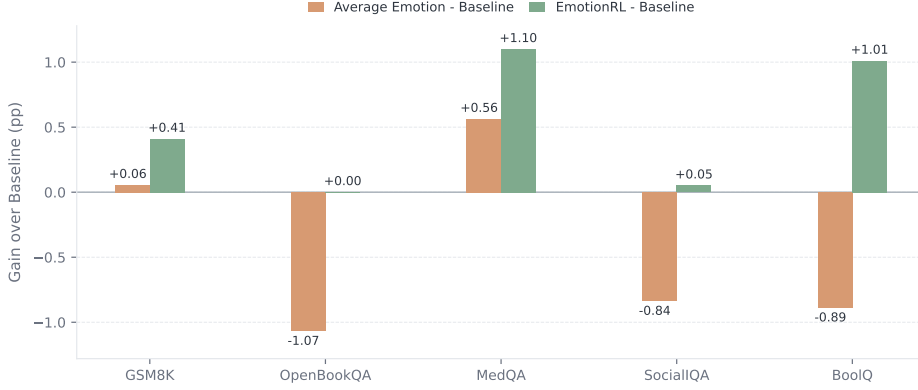


Fig. 6 EmotionRL versus static emotional prompting. Comparison of gains over the no-emotion baseline obtained by averaging across static emotional prompts and by using EmotionRL to select an emotion for each example. Static emotional prompting is inconsistent and can be neutral or harmful on some tasks, whereas EmotionRL is uniformly competitive and often better across the five tasks shown here. This pattern suggests that affective prompting is more useful when treated as an input-conditioned control decision rather than as a single global template.

removes the clear negative average effect observed for several datasets under static prompting and turns emotional framing into a more reliable intervention.

This comparison changes the interpretation of the near-zero averages in Figure 3. The issue is not that emotion is irrelevant, but that a single fixed emotion is too coarse for a heterogeneous benchmark. Once emotion selection is conditioned on the input, the residual task- and instance-level structure becomes exploitable. EmotionRL therefore suggests that affective prompting is better understood as an adaptive control problem than as a universal prompt template.

4 Discussion

Our experiments support a more conservative view of emotional prompting than is sometimes suggested by isolated positive examples. On standard accuracy-based benchmarks, fixed emotional prefixes are usually too weak and too heterogeneous to serve as a dependable performance intervention. For most tasks, the model’s underlying reasoning and knowledge dominate the small perturbation introduced by affective wording.

A second lesson is that aggregate averages can hide the structure of the effect. Near-zero mean change does not imply complete invariance. Instead, it often reflects cancellation between small gains and small losses that depend on both the task and the backbone model. The larger spread on SocialIQA suggests that emotional context matters most when the benchmark itself is socially grounded, whereas more constrained tasks such as arithmetic and medical multiple-choice question answering are comparatively robust.

The intensity and human-versus-LLM studies sharpen this interpretation. Increasing emotional salience on MedQA-US introduces only a mild additional shift,

indicating that stronger emotion changes magnitude but not regime. Likewise, the close agreement between human-written and LLM-generated prefixes shows that the main findings are not artifacts of a particular prefix-generation pipeline. Together, these results suggest that the relevant variable is the fit between emotional framing and the input, not the specific source of the text.

EmotionRL provides the clearest evidence that the observed heterogeneity is not merely noise. A fixed emotion is too coarse to help reliably, but an input-conditioned policy can recover modest and more consistent gains. This suggests a useful reframing: the key question is not whether one emotion works in general, but when a particular emotional framing is appropriate for a particular input. In this sense, affective prompting is better viewed as a routing problem than as a universally beneficial heuristic.

Our study also has important limitations. We focus on short prefixes, single-turn prompting, and accuracy-oriented benchmarks. Larger or qualitatively different effects may emerge in multi-turn interactions, open-ended generation, safety-sensitive dialogue, or evaluations where calibration, style, or perceived empathy matter as much as correctness. Extending adaptive emotion selection to those settings is a natural direction for future work.

5 Conclusion

In this work, we study how emotional tone in queries affects large language models across reasoning, medical question answering, reading comprehension, commonsense QA, and social inference. Across benchmarks, fixed emotional prefixes produce mostly small accuracy shifts, with the clearest sensitivity appearing in socially grounded tasks. Stronger emotional wording leads to only mild additional change, and human-written prefixes reproduce the same qualitative pattern as LLM-generated ones.

The central takeaway is twofold. First, static emotional prompting is not a reliable general-purpose method for improving model performance. Second, emotional framing is not irrelevant: when it is selected adaptively with EmotionRL, it yields more consistent gains than static emotional prompting. Taken together, these findings suggest that affective phrasing is usually a mild perturbation, but its residual signal can be exploited when emotion selection is matched to the input. We hope this provides a cleaner empirical foundation for future work on affective prompting, robust evaluation, and adaptive control of LLM behavior.

Appendix A Emotion prefix generation prompts

We generate one-sentence emotional prefixes with GPT-4o. The **system** message sets the role and safety constraints; the **user** message supplies the target emotion label and the task question text. Replace the placeholder text in the boxes below with the exact strings, temperature, and any few-shot examples used in your generation run.

Prompt Structure

System message.

Generate one **distinct emotional sentence** tailored to appear **before** a given question.

Core rules

1. **Fluency & Logic:** Sentences must sound natural and human, without changing or answering the question itself.
2. **Integrity:** Do not modify the question's numbers, scope, difficulty, or factual details.
3. **Perspective:** The emotion reflects the person's feeling toward **you** (the assistant) solving the problem; they will not help.
4. **Emotion Accuracy:** Each sentence must vividly and truthfully express the assigned emotion, even if rude or unfiltered.
5. **Grammar:** Sentences must be syntactically correct. Typos and broken grammar are not allowed. Exclamation points and similar punctuation may be used to intensify emotion.

Language intensity (raw emotion is allowed)

- Emotionally raw, frustrated, or sarcastic language is allowed if it fits the emotion.
- Rude words, slang, or mild expletives (e.g., "damn," "hell," "you dumb ass") are allowed if they enhance emotional realism.
- Not allowed: targeted hate speech, slurs, or explicit sexual language.
- Anger should feel heated, not hateful; disgust may sound grossed-out or exasperated; fear may sound panicked; happiness may sound ecstatic.

Position rules

- **prepended_sentence** appears before the question.

Form

- Exactly one complete sentence.
- Length: 5–8 words.
- Ends with a single punctuation mark (., ?, or !).
- Use first-person or neutral phrasing.
- Avoid moralizing or instructive words such as "obviously," "just," or "clearly."

User message.